Microsoft Excel

Chapter 2

Advanced Formulas & Functions

Chapter 2

Advanced Formulas & Functions

Formulas are the engine of Microsoft Excel. They transform a plain spreadsheet into a powerful tool that can calculate, analyze, compare, and summarize data automatically. In this chapter, you will begin with the most commonly used formulas, then progressively build your skills toward advanced functions used by professionals in every industry.

📘 What You Will Learn in This Chapter

Revision of basic formulas: SUM, IF, and VLOOKUP
Writing IF formulas with multiple conditions (Nested IF and IFS)
Using logical functions: AND, OR, and NOT
Looking up data with VLOOKUP, HLOOKUP, and XLOOKUP
Using INDEX & MATCH as a powerful alternative to VLOOKUP
Cleaning and working with text using LEFT, RIGHT, MID, LEN, CONCAT, and TRIM

2.1 Revision of Basic Formulas

Before we move to advanced formulas, it is essential to be confident with three foundational formulas that form the backbone of Excel work. These are SUM, IF, and VLOOKUP. Even in professional environments, these three are used daily. In this section, we will revisit them clearly and completely.

2.1.1 The SUM Formula

The SUM formula is used to add up a range of numbers. Instead of manually typing each number into a calculator, you select a range and Excel adds them all instantly.

Syntax

=SUM(number1, number2, ...) or =SUM(A1:A10)
Adds all values in the selected range or list of values

Practical Example — Monthly Sales Summary

Imagine you are an accounts executive at a firm. You have recorded the monthly sales figures for six months in column B. You want to find the total sales for the half-year.

Cell	Month	Sales Amount (₹)
B2	January	1,20,000
B3	February	98,500
B4	March	1,45,000
B5	April	1,10,000
B6	May	1,32,000
B7	June	1,08,000
B8	TOTAL	=SUM(B2:B7)

📝 **Result Explained**

In cell B8, you type: =SUM(B2:B7)
Excel adds all values from B2 to B7 and displays the result: ₹7,13,500
If any value changes, the total updates automatically — no need to recalculate manually.

SUM with Non-Continuous Ranges

Sometimes your data is not in a single continuous column. You can still sum multiple separate ranges by separating them with commas.

=SUM(B2:B4, B7:B9, B12)
Adds values from three separate ranges and one individual cell

💡 **Pro Tip**

You can use AutoSum (shortcut: Alt + =) to instantly insert a SUM formula for the range directly above or to the left of your selected cell.

2.1.2 The IF Formula

The IF formula checks whether a condition is TRUE or FALSE, and returns different results depending on the answer. It is one of the most widely used formulas in professional Excel work — used in everything from grading students to tracking inventory.

Syntax

=IF(logical_test, value_if_true, value_if_false)
Tests a condition and returns one value if TRUE, another if FALSE

Think of the IF formula as a question-and-answer system:

- IF the condition is TRUE ⟶ return this value
- IF the condition is FALSE ⟶ return that value

Practical Example — Student Pass or Fail

You are a teacher and have recorded student marks in column B. If a student scores 40 or above, they pass. Otherwise, they fail. Instead of checking each row manually, you write one IF formula:

Cell	Student Name	Marks (Out of 100)	Result (Formula)
A2	Arjun Mehta	72	=IF(C2>=40,"Pass","Fail") ⟶ Pass
A3	Priya Shah	38	=IF(C3>=40,"Pass","Fail") ⟶ Fail
A4	Rahul Gupta	55	=IF(C4>=40,"Pass","Fail") ⟶ Pass
A5	Sneha Patel	40	=IF(C5>=40,"Pass","Fail") ⟶ Pass

🔎 **Analogy — IF Formula as a Decision Gate**

Think of the IF formula as a checkpoint at the entrance of a building.

> The checkpoint rule is: "Is this person on the approved list?"
> If YES → they enter.
> If NO → they are redirected.
> The IF formula works the same way — it checks a condition and directs the output accordingly.

2.1.3 The VLOOKUP Formula

VLOOKUP stands for Vertical Lookup. It is used to search for a value in the leftmost column of a table and retrieve a corresponding value from another column in the same row. It is like a search engine inside your spreadsheet.

Syntax

> **=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])**
> *Searches for a value in the first column and returns data from a specified column*

Parameter Breakdown:

Parameter	What It Means	Example
lookup_value	The value you are searching for	Employee ID: E104
table_array	The full data table to search in	A2:D20 (your employee table)
col_index_num	Which column number to return data from	3 (3rd column = Department)
range_lookup	FALSE = exact match, TRUE = approximate match	FALSE (always use for exact data)

Practical Example — Employee Information Lookup

Your HR department maintains a master employee list. When you type an Employee ID, you want Excel to automatically retrieve the employee's name and department. This is a classic use case for VLOOKUP.

Employee ID	Name	Department	Salary (₹)
E101	Anita Verma	Finance	45,000
E102	Rohit Desai	Marketing	52,000
E103	Meera Nair	Operations	48,000
E104	Karan Joshi	HR	42,000
E105	Divya Rao	Finance	55,000

To look up the department of Employee E104:

=VLOOKUP("E104", A2:D6, 3, FALSE)

Returns: "HR" — the value from the 3rd column (Department) in the row where E104 is found

⚠️ Important Rule

The lookup value must always be in the FIRST (leftmost) column of your table_array.

VLOOKUP can only search to the RIGHT. If your data is to the left of the lookup column, use INDEX & MATCH instead (covered in Section 2.5).

Always use FALSE for exact match unless you have a specific reason for approximate matching.

2.2 IF with Multiple Conditions — Nested IF and IFS

In real-world situations, decisions often have more than two outcomes. For example, grading a student is not just Pass or Fail — it could be A, B, C, D, or Fail based on different score ranges. Excel provides two ways to handle this: Nested IF and the IFS function.

2.2.1 Nested IF

A Nested IF is an IF formula placed inside another IF formula. Each IF handles one condition, and if that condition is not met, it passes control to the next IF.

Syntax

=IF(condition1, result1, IF(condition2, result2, IF(condition3, result3, default_result)))

Multiple conditions evaluated in sequence — first TRUE match wins

Practical Example — Employee Performance Rating

Your company conducts annual performance reviews. Employees are rated based on their performance score:

Score Range	Rating
90 and above	Outstanding
75 to 89	Exceeds Expectations
60 to 74	Meets Expectations
45 to 59	Needs Improvement
Below 45	Unsatisfactory

The formula to assign ratings automatically:

=IF(B2>=90,"Outstanding", IF(B2>=75,"Exceeds Expectations", IF(B2>=60,"Meets Expectations", IF(B2>=45,"Needs Improvement","Unsatisfactory"))))

Cell B2 contains the employee's performance score

Employee	Score	Rating (Formula Result)
Arjun Mehta	92	Outstanding
Priya Shah	78	Exceeds Expectations
Rahul Gupta	63	Meets Expectations
Sneha Patel	50	Needs Improvement

Employee	Score	Rating (Formula Result)
Vikas Kumar	38	Unsatisfactory

> ⚠️ **Watch Out — Order Matters in Nested IF**
> Always start with the HIGHEST condition and work downward.
> If you start with IF(B2>=45,...) first, a score of 90 would be caught by this condition and incorrectly rated.
> Excel evaluates each condition in sequence and stops at the FIRST condition that is TRUE.

2.2.2 The IFS Function

The IFS function was introduced in Excel 2019 and Microsoft 365. It is a cleaner, easier-to-read alternative to Nested IF. Instead of nesting formulas inside each other, IFS evaluates multiple conditions one after another in a single, flat formula.

Syntax

=IFS(condition1, result1, condition2, result2, condition3, result3, ...)

Evaluates conditions in order — returns the result of the first TRUE condition

Practical Example — The Same Performance Rating Using IFS

The same performance rating problem from above, written using IFS — notice how much more readable it becomes:

=IFS(B2>=90,"Outstanding", B2>=75,"Exceeds Expectations", B2>=60,"Meets Expectations", B2>=45,"Needs Improvement", TRUE,"Unsatisfactory")

The final TRUE acts as a catch-all default (equivalent to the "else" in Nested IF)

> 💡 **Pro Tip — TRUE as a Default in IFS**
> Always end your IFS formula with TRUE, "Your Default Text" to catch any values that do not match any previous condition.
> This prevents the formula from returning an error when no condition is met.

Feature	Nested IF	IFS Function
Readability	Gets harder to read as conditions increase	Flat and easy to read
Maximum conditions	Up to 64 levels (impractical)	Up to 127 conditions
Default (else) value	Last nested IF result	Use TRUE as final condition

Feature	Nested IF	IFS Function
Excel version required	All versions	Excel 2019 / Microsoft 365 only

2.3 Logical Functions — AND, OR, NOT

Logical functions allow you to test multiple conditions at the same time. On their own, they return TRUE or FALSE. Their real power comes when combined with IF, IFS, and other formulas to create intelligent decision-making formulas.

2.3.1 The AND Function

The AND function checks whether ALL of the specified conditions are TRUE. If even one condition is FALSE, the entire AND result is FALSE.

Syntax

=AND(condition1, condition2, condition3, ...)
Returns TRUE only when ALL conditions are TRUE

🔎 Analogy — AND as a Multi-Key Lock

Imagine a safety deposit box that requires THREE keys simultaneously to open.
If any one key is missing, the box stays locked.
AND works the same way — every condition must be TRUE for the result to be TRUE.

Practical Example — Loan Eligibility Check

A bank approves a personal loan only when all three criteria are met:

- Monthly income is ₹50,000 or more
- Credit score is 700 or above
- Age is between 21 and 60

=IF(AND(B2>=50000, C2>=700, D2>=21, D2<=60), "Approved", "Rejected")
B2 = Monthly Income, C2 = Credit Score, D2 = Age

Applicant	Income (₹)	Credit Score	Age	Result
Ramesh Gupta	65,000	720	34	Approved
Sonal Mehta	48,000	750	29	Rejected
Arun Sharma	55,000	680	42	Rejected
Pooja Rao	70,000	710	27	Approved

2.3.2 The OR Function

The OR function checks whether AT LEAST ONE of the specified conditions is TRUE. If any single condition is TRUE, the result is TRUE. Only when ALL conditions are FALSE does OR return FALSE.

Syntax

=OR(condition1, condition2, condition3, ...)
Returns TRUE when at least ONE condition is TRUE

🔎 Analogy — OR as Any One Key

Imagine a door that can be opened with ANY one of three different keys.
If you have at least one key, you can enter.
OR works the same way — if any condition is TRUE, the result is TRUE.

Practical Example — Scholarship Eligibility

A college offers a merit scholarship to students who qualify under at least ONE of the following criteria:

- Academic score is 85% or above, OR
- Has won a national-level competition, OR
- Belongs to a low-income household (annual income below ₹2,50,000)

=IF(OR(B2>=85, C2="Yes", D2<250000), "Eligible", "Not Eligible")
B2 = Academic Score, C2 = National Award (Yes/No), D2 = Annual Family Income

Student	Score (%)	National Award	Family Income (₹)	Scholarship
Anika Singh	88	No	4,20,000	Eligible
Dev Patel	72	Yes	5,00,000	Eligible
Riya Joshi	70	No	2,10,000	Eligible
Nikhil Rao	65	No	5,50,000	Not Eligible

2.3.3 The NOT Function

The NOT function simply reverses a logical value. If the result would be TRUE, NOT makes it FALSE. If it would be FALSE, NOT makes it TRUE. It is used to check for the absence of a condition.

Syntax

=NOT(condition)
Reverses the logical result — TRUE becomes FALSE and FALSE becomes TRUE

Practical Example — Flagging Incomplete Submissions

In an HR system, you want to flag employees who have NOT submitted their annual declaration form. The submission status is recorded as Yes or No.

=IF(NOT(B2="Yes"), "Reminder Required", "Submitted")
B2 contains "Yes" or "No" based on whether the form was submitted

Employee	Form Submitted	Status
Anita Verma	Yes	Submitted
Rohit Desai	No	Reminder Required
Meera Nair	Yes	Submitted
Karan Joshi	No	Reminder Required

💡 **Combining AND, OR, and NOT with IF**

You can combine these functions: =IF(AND(OR(A1>10, B1>10), NOT(C1="Excluded")), "Include", "Exclude")
Build the inner logic first, test it alone, then wrap it in the IF formula.

2.4 Lookup Functions — VLOOKUP, HLOOKUP, and XLOOKUP

Lookup functions are essential when working with large datasets. They allow you to search for a specific value in a table and automatically retrieve related information. Understanding all three variants gives you the flexibility to work with any data layout.

2.4.1 VLOOKUP — A Deeper Look

You were introduced to VLOOKUP in Section 2.1. Here, we explore it further with common professional scenarios and important limitations to be aware of.

Practical Example — Product Price Lookup

You are working in a retail company. You have a product catalogue with Product Codes and Prices. On a separate order form, when a salesperson enters a Product Code, the price should populate automatically.

Product Code	Product Name	Category	Price (₹)
P001	Wireless Mouse	Peripherals	850
P002	USB Keyboard	Peripherals	1,200
P003	15-inch Monitor	Display	12,500
P004	Laptop Stand	Accessories	1,800
P005	HDMI Cable	Accessories	450

On the order form, when Product Code is entered in cell H2, the price auto-fills:

=VLOOKUP(H2, A2:D6, 4, FALSE)

Searches for H2 in column A, returns value from column 4 (Price)

Common VLOOKUP Errors

Error	Reason	Solution
#N/A	Lookup value not found in the table	Check spelling, spacing, or use IFERROR
#REF!	Column index number exceeds table width	Reduce col_index_num to match actual columns
#VALUE!	Non-numeric col_index_num entered	Make sure col_index_num is a whole number
Wrong result	Using TRUE (approximate) instead of FALSE	Always use FALSE for exact matching

> 💡 **Wrapping VLOOKUP with IFERROR**
> Use IFERROR to show a friendly message instead of an error: =IFERROR(VLOOKUP(H2, A2:D6, 4, FALSE), "Product not found")
> This is considered best practice in professional spreadsheet design.

2.4.2 HLOOKUP — Horizontal Lookup

HLOOKUP works exactly like VLOOKUP, except it searches horizontally across the top row of a table (instead of vertically down the first column). Use HLOOKUP when your lookup keys are arranged in a row rather than a column.

Syntax

> **=HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup])**
> *Searches across the top row and returns data from a specified row number*

Practical Example — Quarterly Target Lookup

Your company stores quarterly sales targets in a horizontal table — each quarter is a column header. You want to look up the target for a specific quarter.

Region / Quarter	Q1	Q2	Q3	Q4
North Zone	8,00,000	9,50,000	10,00,000	12,00,000
South Zone	7,50,000	8,00,000	8,50,000	10,50,000

> **=HLOOKUP("Q3", B1:E3, 2, FALSE)**
> *Returns the Q3 target for North Zone: ₹10,00,000 (Row 2 of the table)*

Feature	VLOOKUP	HLOOKUP
Search direction	Searches DOWN a column	Searches ACROSS a row
Lookup keys location	First (leftmost) column	First (top) row
Returns data from	A column to the RIGHT	A row BELOW the lookup row
Best used when	Data is arranged in rows	Data is arranged in columns

2.4.3 XLOOKUP — Introduction

XLOOKUP is the modern replacement for both VLOOKUP and HLOOKUP, available in Excel 2021 and Microsoft 365. It resolves many of the limitations of the older functions and offers significantly more flexibility.

Syntax

=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode], [search_mode])

Searches any range and returns values from any corresponding range

Parameter	Description
lookup_value	The value you are searching for
lookup_array	The column or row to search in (not the full table)
return_array	The column or row from which to return a result
if_not_found	Optional: What to display if nothing is found (e.g., "Not Found")
match_mode	0 = exact match (default), -1 = next smaller, 1 = next larger
search_mode	1 = first to last (default), -1 = last to first

Practical Example — Employee Lookup with XLOOKUP

Using the same employee table from Section 2.1, XLOOKUP retrieves the department for a given Employee ID:

=XLOOKUP("E104", A2:A6, C2:C6, "Not Found")

Searches for E104 in column A, returns value from column C (Department). If not found: "Not Found"

📝 Why XLOOKUP Is Better Than VLOOKUP

Can look LEFT — no restriction on which side the return column is on.
Does not need a column number — you directly select the return range.
Built-in 'if not found' parameter — no need for a separate IFERROR wrapper.
Can search bottom-to-top (useful for finding the most recent entry).
Works with both vertical and horizontal data natively.

2.5 INDEX & MATCH — The Professional Alternative to VLOOKUP

INDEX and MATCH are two separate functions that, when combined, form one of the most powerful and flexible lookup tools in Excel. While VLOOKUP is limited to searching the leftmost column and returning data to the right, INDEX & MATCH can search any column and return data from any direction.

2.5.1 Understanding the INDEX Function

The INDEX function returns the value of a cell in a given row and column position within a specified range. Think of it as reading a specific cell using coordinates.

Syntax

=INDEX(array, row_num, [col_num])
Returns the value at the intersection of the specified row and column within a range

Simple Example

	Column 1 (A)	Column 2 (B)	Column 3 (C)
Row 1	Apple	Red	₹80
Row 2	Banana	Yellow	₹40
Row 3	Mango	Orange	₹120

=INDEX(A1:C3, 2, 3)
Returns the value in Row 2, Column 3 → Result: ₹40

🔎 **Analogy — INDEX as a Grid Reference**

Think of a spreadsheet like a map grid.
INDEX(A1:C3, 2, 3) means: Go to Row 2, Column 3 of this grid and tell me what is there.
Just like saying Latitude 2, Longitude 3 on a map.

2.5.2 Understanding the MATCH Function

The MATCH function searches for a value in a row or column and returns its position number. It does not return the value itself — it returns where the value is located.

Syntax

=MATCH(lookup_value, lookup_array, [match_type])

Returns the position (row or column number) of the lookup value within the array

match_type	Behaviour
0	Exact match (most commonly used)
1	Finds the largest value less than or equal to lookup_value (requires sorted data)
-1	Finds the smallest value greater than or equal to lookup_value (requires sorted data)

Simple Example

=MATCH("Banana", A1:A3, 0)

Returns 2 — because Banana is in the 2nd position in the range A1:A3

2.5.3 Combining INDEX and MATCH

The real power comes when you nest MATCH inside INDEX. MATCH finds the row position of your search value, and INDEX uses that position to retrieve the corresponding data. This combination overcomes all the limitations of VLOOKUP.

The Formula Structure

=INDEX(return_column, MATCH(lookup_value, search_column, 0))

MATCH finds the row number; INDEX uses it to retrieve the value from any column

Practical Example — Product Lookup with Left-Side Return

You have a product table where the Product Code is in Column C and the Product Name is in Column A (to the LEFT of the code). VLOOKUP cannot retrieve data from the left. INDEX & MATCH can.

Product Name (A)	Category (B)	Product Code (C)	Price (D)
Wireless Mouse	Peripherals	P001	₹850
USB Keyboard	Peripherals	P002	₹1,200
15-inch Monitor	Display	P003	₹12,500
Laptop Stand	Accessories	P004	₹1,800

To find the Product Name when you know the Product Code is P003:

=INDEX(A2:A5, MATCH("P003", C2:C5, 0))
MATCH finds P003 at position 3; INDEX returns A4 (row 3 of column A): "15-inch Monitor"

Feature	VLOOKUP	INDEX & MATCH
Search column restriction	Must be the leftmost column	Can search any column
Return direction	Can only return to the RIGHT	Can return from any direction
Column number sensitivity	Breaks if columns are inserted	Stable — uses column references
Performance on large data	Slower on very large datasets	Generally faster and more efficient
Complexity	Simpler to write	Slightly more complex but highly flexible

2.6 Text Functions — LEFT, RIGHT, MID, LEN, CONCAT, TRIM

In professional data work, you will frequently encounter text that needs to be cleaned, split, combined, or formatted. Excel's text functions are designed to handle exactly these situations — from extracting portions of a product code to combining first and last names into a full name.

2.6.1 LEFT — Extract from the Beginning

The LEFT function extracts a specified number of characters from the beginning (left side) of a text string.

Syntax

=LEFT(text, num_chars)
Returns the leftmost N characters from the text

Practical Example — Extracting Region Code from Employee ID

Your company uses Employee IDs in the format: REG-EMP-001 where the first 3 characters represent the region. You want to extract only the region code.

Employee ID (A)	Region Code Formula	Result
MUM-EMP-001	=LEFT(A2, 3)	MUM
DEL-EMP-042	=LEFT(A3, 3)	DEL
BLR-EMP-019	=LEFT(A4, 3)	BLR

2.6.2 RIGHT — Extract from the End

The RIGHT function extracts a specified number of characters from the end (right side) of a text string.

Syntax

=RIGHT(text, num_chars)
Returns the rightmost N characters from the text

Practical Example — Extracting Invoice Number

Invoice codes are formatted as INV-2024-0051. You want to extract only the numeric portion at the end.

=RIGHT(A2, 4)
From "INV-2024-0051", extracts the last 4 characters: "0051"

Invoice Code (A)	Formula	Extracted Number
INV-2024-0051	=RIGHT(A2, 4)	0051
INV-2024-0128	=RIGHT(A3, 4)	0128
INV-2024-0003	=RIGHT(A4, 4)	0003

2.6.3 MID — Extract from the Middle

The MID function extracts a specified number of characters from a specific position within a text string. You define where to start and how many characters to extract.

Syntax

=MID(text, start_num, num_chars)
Extracts num_chars characters starting at position start_num

Practical Example — Extracting Year from Invoice Code

Using the same invoice code format INV-2024-0051, you want to extract the year "2024" which starts at position 5 and is 4 characters long.

=MID(A2, 5, 4)
From "INV-2024-0051", starts at position 5, takes 4 characters: "2024"

Invoice Code	=LEFT(A,3)	=MID(A,5,4)	=RIGHT(A,4)
INV-2024-0051	INV	2024	0051
INV-2025-0128	INV	2025	0128

🔎 How to Find the Start Position
Count characters from left: Position 1 = first character.
For "INV-2024-0051": I=1, N=2, V=3, -=4, 2=5 ⟶ Year starts at position 5.
Alternatively, use the FIND or SEARCH function to locate a specific character automatically.

2.6.4 LEN — Count Characters

The LEN function counts the total number of characters in a text string, including spaces. It is often used in combination with other text functions to make them dynamic.

Syntax

=LEN(text)
Returns the total number of characters in the text, including spaces

Practical Example — Validating PIN Code Length

Your data entry form should accept only 6-digit PIN codes. You want to flag any entry that does not have exactly 6 characters.

=IF(LEN(A2)=6, "Valid", "Invalid — Check PIN")
A2 contains the entered PIN code

Entered PIN (A)	=LEN(A)	Validation Result
400001	6	Valid
38002	5	Invalid — Check PIN
1100011	7	Invalid — Check PIN
560094	6	Valid

💡 **Pro Tip — Using LEN with RIGHT and MID**

LEN becomes powerful when combined with other functions:
=RIGHT(A2, LEN(A2)-4) extracts everything after the first 4 characters — dynamically, regardless of total length.
This technique is used when text lengths vary across rows.

2.6.5 CONCAT — Combine Text

The CONCAT function joins (concatenates) two or more text strings into a single string. It is the modern replacement for the older CONCATENATE function. You can also use the & operator as a quick alternative.

Syntax

=CONCAT(text1, text2, text3, ...) or =text1 & " " & text2
Joins multiple text values into one continuous string

Practical Example — Creating Full Names from First and Last Name Columns

Your employee database stores First Name and Last Name in separate columns. You want to create a Full Name column for reports and communications.

First Name (A)	Last Name (B)	Full Name Formula	Result
Anita	Verma	=CONCAT(A2," ",B2)	Anita Verma
Rohit	Desai	=A3&" "&B3	Rohit Desai
Meera	Nair	=CONCAT(A4," ",B4)	Meera Nair

Practical Example — Generating Employee IDs

You need to auto-generate Employee IDs by combining a department code, a hyphen, and a serial number:

=CONCAT(B2, "-", TEXT(C2,"000"))
B2 = Department Code (e.g., "HR"), C2 = Serial Number (e.g., 4) → Result: "HR-004"

📝 **CONCAT vs CONCATENATE vs &**

CONCATENATE is the old function (still works but not recommended for new work).
CONCAT is the modern replacement — cleaner and supports ranges.
& operator is the quickest for simple joining: =A1&" "&B1
TEXTJOIN (not covered here) is ideal when joining many cells with the same separator.

2.6.6 TRIM — Remove Extra Spaces

The TRIM function removes all leading spaces (before the text), trailing spaces (after the text), and extra spaces between words — leaving only single spaces between words. This is extremely important when working with data imported from other systems.

Syntax

=TRIM(text)
Removes leading, trailing, and extra internal spaces from text

🔎 **Why TRIM Is Critical in Professional Work**

When data is imported from databases, ERPs, or CSV files, entries often contain hidden extra spaces.
A cell containing " Anita Verma " looks identical to "Anita Verma" visually, but VLOOKUP and MATCH will fail to find it because the spaces make them different strings.
TRIM ensures your data is clean before lookups and comparisons.

Practical Example — Cleaning Imported Customer Names

Raw Imported Name (A)	After TRIM (=TRIM(A))	Characters (LEN)
Anita Verma	Anita Verma	16 → 11
Rohit Desai	Rohit Desai	13 → 11
Meera Nair	Meera Nair	10 → 10 (no change)

> 💡 **Pro Tip — Combine TRIM with Other Functions**
> Always apply TRIM when building lookup formulas on imported data:
> =VLOOKUP(TRIM(H2), A2:D6, 4, FALSE)
> This ensures even if the search value has accidental spaces, the lookup still works correctly.

2.6.7 Text Functions — Quick Reference Summary

The following table summarises all text functions covered in this section for easy reference:

Function	Purpose	Example	Result
LEFT(text, n)	First n characters	=LEFT("INV-2024", 3)	INV
RIGHT(text, n)	Last n characters	=RIGHT("INV-2024", 4)	2024
MID(text, start, n)	n characters from position start	=MID("INV-2024", 5, 4)	2024
LEN(text)	Total character count	=LEN("Mumbai")	6
CONCAT(t1, t2, ...)	Join multiple texts	=CONCAT("A","-","001")	A-001
TRIM(text)	Remove extra spaces	=TRIM(" Hello ")	Hello

Chapter Summary

This chapter covered the core formulas and functions that form the foundation of advanced Excel usage. Here is a consolidated review of everything learned:

Key Concepts Covered

Section 2.1 — Basic Formulas

- SUM adds up a range of numbers — essential for any financial or numerical summary.
- IF evaluates a condition and returns different values based on TRUE or FALSE.
- VLOOKUP searches a table vertically and retrieves data from a specified column to the right.

Section 2.2 — IF with Multiple Conditions

- Nested IF handles scenarios with more than two outcomes by placing IFs inside IFs.
- IFS provides a cleaner syntax for multiple conditions (Excel 2019 / Microsoft 365).

Section 2.3 — Logical Functions

- AND returns TRUE only when all conditions are TRUE (like a multi-requirement gate).
- OR returns TRUE when at least one condition is TRUE.
- NOT reverses a logical result — useful for checking the absence of a condition.

Section 2.4 — Lookup Functions

- VLOOKUP searches the first column and returns data from a column to the right.
- HLOOKUP searches the first row and returns data from a row below.
- XLOOKUP is the modern replacement — searches any range, returns any range, and includes a built-in not-found message.

Section 2.5 — INDEX & MATCH

- INDEX retrieves a value based on its row and column position in a range.
- MATCH finds the position of a value in a range.
- Combined, they form a flexible lookup that can search and return data in any direction.

Section 2.6 — Text Functions

- LEFT, RIGHT, MID extract specific portions of text.
- LEN counts the total number of characters.
- CONCAT joins multiple text values into one.
- TRIM removes all unnecessary spaces — critical for clean data and reliable lookups.

📌 Professional Best Practices to Remember

Always use FALSE in VLOOKUP and XLOOKUP for exact matching in business data.
Use IFERROR or the if_not_found parameter to handle missing data gracefully.
Apply TRIM to data imported from external sources before running any lookups.
Use IFS instead of deeply Nested IFs for readability and maintainability.
Prefer INDEX & MATCH or XLOOKUP over VLOOKUP in professional-grade spreadsheet design.